

James V. Raj

Ph.D. Candidate, Department of Physics and Astronomy

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EDUCATION

University of California, Los Angeles

September 2022 to present

Ph.D. in Physics

Expected June 2027

Advisor: [Mason A. Porter](#). Advanced to candidacy March 2025.

California Polytechnic State University, San Luis Obispo

September 2017 to June 2021

B.S. in Physics

RESEARCH INTERESTS

Networks that arise from physical systems; hyperuniform point patterns; scale-rich metamaterials; network science; physical learning with local learning rules; topological data analysis and persistent homology; statistical physics.

PUBLICATIONS

Journal articles

R. Sengupta, M. D. Tikekar, **J. V. Raj**, K. T. Delaney, M. C. Villet, and G. H. Fredrickson, “Phase-field simulations of morphology development in reactive polymer blending,” *J. Rheol.* **67**, 1 (2023). doi:10.1122/8.0000523

Preprints

J. V. Raj, X. Sun, C. E. Maher, K. A. Newhall, and M. A. Porter, “Local geometric and transport properties of networks that are generated from hyperuniform point patterns,” [arXiv:2511.21082](#) (2025); in revision at *Phys. Rev. E*. Code: github.com/DMREF-networks/HuPPI-Network-Analysis.

Manuscripts in preparation

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “Size-resolved H_0 persistence as an exact coalescence observable,” in preparation (2026).

J. V. Raj, C. E. Maher, K. A. Newhall, and M. A. Porter, “A directional transport surrogate for stiffness in disordered beam networks,” in preparation (2026).

J. V. Raj, L. Gyllingberg, P. B. Pietrzkowski, M. Stern, and M. A. Porter, “Network structure controls how quickly local learning opens spectral gaps in spring networks,” in preparation (2026).

TALKS AND PRESENTATIONS

“Network structure shapes accessible solutions in physical learning,” contributed talk, Conference on Complex Systems (CCS 2026), Binghamton University, Binghamton, NY, October 2026.

“Topological and transport properties of spatially embedded networks generated by hyperuniform point patterns,” contributed talk, APS March Meeting, Anaheim, CA, March 2025.

“Understanding how the glass transition concentration affects polymer membrane microstructure,” talk, Future Leaders in Advanced Materials Summer Symposium, August 2020.

“Electrical current characteristics of simulated carbon nanotube network field-effect transistors,” poster, APS Far West Section Meeting, October 2019.

RESEARCH EXPERIENCE

Graduate Student Researcher, UCLA January 2024 to present
Advisor: Mason A. Porter, Department of Mathematics

Graduate Student Researcher, UCLA June 2022 to June 2023
Advisor: Shenshen Wang, Department of Physics and Astronomy
Theoretical biophysics of adaptive immunity and evolving populations.

Postbaccalaureate Researcher, UC Santa Barbara June 2021 to June 2022
Materials Research Laboratory; advisor: Glenn H. Fredrickson, Department of Chemical Engineering
Designed and ran phase-field simulations of reactive polymer blending under shear flow on UCSB’s high-performance computing clusters; wrote the Python simulation-input and ParaView analysis scripts.

Summer Research Intern (NSF-funded), UC Santa Barbara June 2020 to September 2020
Advisor: Glenn H. Fredrickson
Simulated polymer membrane formation on high-performance computing clusters; wrote Python tools to quantify pore structure in SEM images and compare it with simulations.

Soft Matter Research Assistant, Cal Poly January 2019 to June 2021
Advisor: Hilary Jacks
Statistical analysis of memory formation in non-Brownian suspensions under cyclic shear, including an algorithm to read multiple stored memories from particle positions.

Nanoelectronics Research Assistant, Cal Poly September 2018 to August 2019
Advisor: Colleen Marlow; includes a 2019 summer research internship
Simulations of carbon nanotube network field-effect transistors: network morphology, percolation, and sensitivity to electrostatic gating.

FELLOWSHIPS, AWARDS, AND FUNDING

Outstanding Teaching Award, Department of Physics and Astronomy, UCLA, for the 2022 to 2023 and 2023 to 2024 academic years.

Graduate research support, NSF DMREF award CMMI-2323343 (UCLA), 2024 to present.

Future Leaders in Advanced Materials Internship Award, 2020.

William and Linda Frost Undergraduate Research Award, Cal Poly, 2019.

TEACHING EXPERIENCE

Teaching Assistant, Department of Physics and Astronomy, UCLA Winter 2023 to Spring 2025
Upper division: Astronomy 115, Introduction to Galactic Astronomy (Winter 2023, Winter 2024); Physics 112, Thermal Physics (Spring 2023); Physics 32, Mathematical Methods (Fall 2023).

Lower division: Physics 1A, Mechanics (Winter 2025); Physics 1B, Oscillations, Waves, Electric and Magnetic Fields (Fall 2024, Spring 2025).

MENTORING AND SERVICE

Graduate mentor, UCLA Marginalized Identities in Physics & Astronomy (MiPA), 2023 to 2025: mentored eight undergraduate students.

Graduate student mentor, 2023 to 2024 and 2024 to 2025 academic years: mentored two graduate students.

Assisted with a referee report for *Interfaces and Free Boundaries*, 2025.

Administrator of the Porter research group's shared Linux compute server, 2025 to present.

Vice President, Sigma Pi Sigma (physics honor society), Cal Poly, August 2020 to June 2021.

COMPUTATIONAL SKILLS

Languages: Python, Bash, $\text{\LaTeX}$, JavaScript/HTML/CSS, Rust (sparse numerical kernels called from Python)

Scientific computing: NumPy, SciPy (sparse linear algebra, eigensolvers), NetworkX, pandas, Matplotlib, Plotly, ParaView; parallel computing with multiprocessing and joblib; GUDHI, Ripser, and persim for persistent homology

Machine learning: PyTorch (CUDA, `torch.compile`, differentiable optimization); LoRA fine-tuning of diffusion models; local deployment of open-weight LLMs with llama.cpp (quantization, speculative decoding, mixture-of-experts offloading)

Systems: batch jobs on high-performance computing clusters; Linux server administration (user accounts, systemd services, networking) for a 96-core, 1 TB RAM GPU server; Git and GitHub; pytest